Status Report: Contrastive Learning for Face Recognition in Anime

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1. Project Overview
The goal of this project is to build a model that can identify anime characters from images by learning stable and discriminative visual representations. Given a query image, the model extracts visual features and retrieves the most likely identity from a reference gallery of known characters. This is a fine-grained recognition task: many characters look extremely similar at first glance, often sharing hair color, eye color, facial structure, or clothing cues, while the same character can also look quite different across scenes because of pose, lighting, cropping, or art style. For that reason, we frame the problem as retrieval rather than standard closed-set classification. A retrieval-based pipeline is more scalable and gives us more flexibility when working with a large and growing set of character identities.

2. Related Work
In our original proposal, we framed the task as a vision-language retrieval problem built around a dual-encoder architecture. After further reading and planning, we refined that idea into a more practical version for the course timeline: image-only representation learning for retrieval, using pretrained visual encoders and nearest-neighbor search. The main intuition is unchanged. We still want to learn an embedding space where images of the same character cluster together and images of different characters are separated. What changed is that we now focus on the visual side directly, instead of building a full image-text system.

This adjustment is also supported by prior work. In particular, the paper by Rios et al. shows that large-scale anime face recognition is feasible and that pretrained transformer-based visual encoders can work well in this domain. More broadly, work on contrastive embedding learning and hard-negative retrieval gives a strong motivation for our current training objective and top-k retrieval evaluation. We also keep FAISS-style nearest-neighbor search from the original proposal, since efficient embedding retrieval is still central to the final system.

Compared with the original proposal, our current direction differs in two important ways:

1. **We moved from multimodal retrieval to image-only retrieval.** This makes the system much easier to implement within the course timeline and lets us focus on the hardest part of the problem: learning strong identity-sensitive visual features.
2. **We expanded the model comparison.** Instead of committing to one encoder family from the beginning, we now plan to compare several pretrained visual backbones under

the same training and evaluation setup. Our current candidates are Facebook DINOv3, Bytedance LVFace, Google ViT, and MahmoodLab UNI.

3. Approach

Our current pipeline has three main stages: feature extraction, embedding learning, and nearest-neighbor retrieval.

Let $x \in \mathcal{X}$ denote an anime face image. A pretrained visual backbone $f_\theta(\cdot)$ maps the input image to a feature vector,

$$h = f_\theta(x) \in \mathbb{R}^d.$$

A projection head $g_\psi(\cdot)$ then maps the backbone features into a normalized embedding space,

$$z = \frac{g_\psi(h)}{\|g_\psi(h)\|_2} \in \mathbb{R}^m.$$

For two images x_i and x_j, we define similarity using cosine similarity,

$$s_{ij} = z_i^\top z_j.$$

During training, we use a contrastive objective so that embeddings from the same character are pulled closer together while embeddings from different characters are pushed apart. A supervised contrastive loss fits this setting naturally:

$$\mathcal{L}\text{supcon} = \sum i = 1^B \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(s_{ip}/\tau)}{\sum_{a \neq i} \exp(s_{ia}/\tau)},$$

where:

- B is the batch size
- $P(i)$ is the set of samples in the batch sharing the same identity as sample i
- τ is a temperature hyperparameter

In practice, we may also include a lightweight classification head during fine-tuning to stabilize training, while retrieval quality remains the primary objective.

At inference time, each reference image is encoded once to produce an embedding. Given a query image X_q, we compute its embedding z_q and retrieve the most similar gallery embeddings

using cosine similarity. The system returns the top-k candidate identities.

This design keeps the central idea of the original proposal—a retrieval-based recognition system—but adapts it into a more direct and manageable face-recognition setting. It also makes backbone comparison much cleaner, which is one of our main goals at this stage.

Re-used libraries / implementations. We are building on standard tooling such as PyTorch, Hugging Face / timm model interfaces, and FAISS for nearest-neighbor search.
Additional code written by our team. Our own implementation work includes anime-face data filtering and subset construction, experiment configuration for the four candidate backbones, training and fine-tuning code, retrieval evaluation scripts for Recall@k and MRR, and SCC-compatible scripts for larger-scale training.

4. Dataset and Metric
This part has changed from the original proposal. The proposal centered on the DAF:re dataset, which is derived from the Danbooru2018 illustration repository. The full dataset contains 561,000 anime character cropped head images and 34,000 character classes. But after revisiting scale, feasibility, and available compute based on the proposal feedback, we decided to work from a larger dataset, particularly the Danbooru dataset, 2021 edition, which consists of 4.9 million images and 162 million tags. While this is still nowhere near the amount of 400 million image-text pairs for CLIP training as suggested in the feedback, this is the largest dataset of anime images with tags as of currently. Many groups have come up with successful classification utilizing this dataset[6][7][8]. From this original dataset, we will derive smaller high-frequency subsets for rapid experimentation and tuning. The reason is straightforward: the full source is extremely large, on the order of terabytes, so running end-to-end experiments on the full data from the beginning would make debugging and model comparison unnecessarily slow. There are also a variety of irrelevant tags, as well as images with not enough tags which should be filtered out.

Our current dataset plan therefore has two stages:

1. **Subset stage for development.** We first construct a filtered subset containing characters with enough examples for stable training and evaluation. This gives us a fast way to validate preprocessing, test model behavior, and compare backbones.
2. **Larger-scale stage for final experiments.** Once the pipeline is stable, we will scale the best-performing setup to a much larger portion of the Danbooru-based dataset on BU SCC.

We keep the retrieval-oriented evaluation framework from the proposal. Our main metrics are:

- **Recall@1, Recall@5, Recall@10:** whether the correct character identity appears in the top-k retrieved results.

- **MRR (Mean Reciprocal Rank):** a ranking-sensitive metric that tells us how high the correct identity appears on average.

For all experiments, we will use a train/validation/test split after filtering, with exact counts determined by the final subset construction. The split protocol will be kept consistent across all backbones so that the comparison is fair.

5. Preliminary Results

At this point, our main progress has been in refining the problem setup, selecting candidate models, and designing a realistic pipeline rather than in producing final quantitative benchmarks.

What we have completed so far:

- We refined the project scope and moved from a broad vision-language retrieval idea to a more focused image-only contrastive retrieval pipeline.
- We identified four candidate pretrained backbones for comparison: DINOv3, LVFace, ViT, and UNI.
- We reviewed prior work, especially evidence that large-scale anime recognition is feasible with Danbooru-derived data and pretrained transformer-style features.
- We adopted a subset-first development strategy because the full dataset is too large for efficient early experimentation.
- We finalized the evaluation framework around Recall@k and MRR so that all candidate backbones can be tested under one consistent protocol.

Because fine-tuning on the filtered subset is our next major milestone, we do not yet report quantitative retrieval scores. We would rather leave this section honest than include numbers that are incomplete or not yet reliable. At the moment, our most meaningful result is that the project has converged on a technically coherent and computationally feasible experimental plan.

The table below summarizes our current implementation status.

Component	**Status**	**Notes**
Problem formulation refinement	Done	Narrowed from dual-encoder proposal to image-only retrieval
Candidate backbone selection	Done	DINOv3, LVFace, ViT, UNI
Dataset scaling plan	Done	Subset-first, then SCC scaling

Component	Status	Notes
Retrieval metric design	Done	Recall@1/5/10 and MRR
Unified training pipeline	In progress	Shared fine-tuning setup across backbones
Large-scale SCC training	Not started	Scheduled after subset validation
Quantitative benchmark table	Not started	Will be added after the first controlled runs

6. Detailed Timeline and Roles

The remaining work is organized so that each teammate owns a meaningful technical component while still contributing to integration and final write-up.

Task	Deadline	Who
Finalize filtered subset construction and preprocessing pipeline	04/11/26	Jiaqiang Zhu, Minh Nguyen
Implement unified training loop and model wrappers for DINOv3 / ViT / UNI / LVFace	04/13/26	Jun Xie
Build retrieval evaluation pipeline (Recall@1/5/10, MRR, gallery/query protocol)	04/14/26	Jiaqiang Zhu
Prepare SCC training scripts and large-scale experiment configuration	04/15/26	Julio Reyess
Run controlled subset experiments for all four backbones	04/17/26	All
Analyze results and select final backbone	04/19/26	Jiaqiang Zhu, Jun Xie
Scale selected setup to a larger BU SCC training run	04/21/26	Julio Reyess, Minh Nguyen

Task	Deadline	Who
Draft final report figures, tables, and discussion	04/23/26	All
Prepare report and presentation	See course schedule	All

7. Preliminary Code

Our codebase already contains the early training and evaluation framework for the project. The public repository link will be added once the README and experiment structure are cleaned up for release.

GITHUB: https://github.com/SoymilkWinsAgain/EC523_Final_Project

References

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